

Ride Booking Analytics Dashboard Using R and Apache Hive

STQD6324 Data Management

Student Information

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1. Introduction

This project develops an interactive Ride Booking Analytics Dashboard using R Shiny. The objective is to demonstrate data management, data integration, data processing, and visualization techniques using multiple approaches.

The project uses a ride-booking dataset containing booking information, vehicle types, booking values, ride distances, cancellation status, ratings, and payment methods.

Three different data processing approaches were implemented:

1. Direct data processing using R.
2. Data processing using Apache Hive and exporting cleaned data through WinSCP.
3. Direct connection between RStudio and Hive using ODBC.

The final output is an interactive dashboard that provides booking analytics, revenue insights, cancellation analysis, and predictive analytics.

2. Project Objectives

The objectives of this project are:

- To manage large ride-booking data efficiently.
- To perform data cleaning and preprocessing.
- To integrate Hive with RStudio.
- To visualize business insights using Shiny Dashboard.
- To develop a cancellation risk prediction model.

3. Dataset Description

The dataset contains ride-booking transaction records.

Main attributes include:

Attribute	Description
booking_id	Booking identifier
customer_id	Customer identifier
Vehicle_Type	Vehicle category
Booking_Status	Booking status
Payment_Method	Payment method
Pickup_Location	Pickup location
drop_location	Destination location
Ride_Distance	Distance travelled
Booking_Value	Booking revenue
Driver_Ratings	Driver rating
Customer_Rating	Customer rating
Is_Cancelled	Cancellation indicator

Dataset Size:

- Total Records: 103,024
- Cancelled Records: 39,057
- Completed Records: 63,967

```

10/20/2019
Time taken: 116.282 seconds, Fetched: 1 row(s)
hive> DESCRIBE bookings_raw;
OK
booking_date      string
booking_time      string
booking_id        string
booking_status    string
customer_id       string
vehicle_type      string
pickup_location   string
drop_location     string
v_tat            string
c_tat            string
canceled_rides_by_customer string
canceled_rides_by_driver  string
incomplete_rides      string
incomplete_rides_reason string
booking_value         string
payment_method        string
ride_distance         string
driver_ratings        string
customer_rating       string
vehicle_images        string
Time taken: 0.514 seconds, Fetched: 20 row(s)

```

4. Methodology

The project was implemented using three different approaches.

4.1 Approach 1: Direct Data Processing Using R

Process Flow

CSV Dataset → RStudio → Data Cleaning → Dashboard

Steps:

1. Import CSV dataset into R.
2. Clean missing values.
3. Create derived variables.
4. Generate dashboard visualizations.
5. Build cancellation prediction model.

Technologies Used:

- R
- Tidyverse
- Lubridate
- Plotly
- DT
- Shiny Dashboard

```

glimpse(df)
Rows: 103,024
Columns: 28
# Date                                <dtm> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, -
# Time                               "14:00:00", "22:20:00", "19:59:00", "03:15:00,-
# Booking_ID                         "CNR715325142", "CNR294042040", "CNR2982357,-
# Booking_Status                     "Canceled by Driver", "Success", "Success", "-
# Customer_ID                        "CID713523", "CID252458", "CID270156", "CID58,-
# Vehicle_Type                       "Prime Sedan", "Bike", "Prime SUV", "EBike",-
# Pickup_Location                    "Tumkur Road", "Magadi Road", "Sahakar Nagar,-
# Drop_Location                      "RT Nagar", "Varthur", "Vijayanagara,-
# V_TAT                              <dbl> NA, 203, 238, NA, 252, 231, 133, NA, NA, NA, NA,-
# C_TAT                              <dbl> NA, 30, 130, NA, 80, 90, 40, NA, NA, NA, 55,-
# Completed_Rides_by_Customer        "Personal", "Driver not moving towards", "Wajayana,-
# Canceled_Rides_by_Driver           "personal & car related issue", "NA, NA, NA, N,-
# Incomplete_Rides                  "NA", "No", "No", "No", "No", "No", "No", "NA, NA,-
# Incomplete_Rides_Reason            "NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,-
# Customer_Value                     444, 18, 386, 384, 822, 173, 140, 189, 189, 189,-
# Payment_Method                    "NA", "Cash", "UPI", "NA", "credit Card", "UPI",-
# Ride_Distance                     <dbl> 0, 13, 40, 0, 45, 41, 49, 0, 0, 0, 46, 18, 8,-
# Driver_Ratings                     <dbl> 4.0, 4.1, 4.2, 4.0, 4.0, 3.4, 3.2, 4.0, 4.0,-
# Customer_Rating                    <dbl> 4.0, 4.0, 3.8, 4.0, 4.0, 3.0, 4.6, 4.0, 4.0, 4.0, 4.0,-
# Year                              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,-
# Month                             <ord> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,-
# Day                              <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,-
# Weekday                           <ord> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,-
# Hour                              <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,-
# Is_Canceled                       <dbl> 1, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0,-
# Success                           <dbl> 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0, 1,-
# Revenue_Category                   "Medium", "Low", "medium", "Medium", "High", "-
# Distance_Category                 <chr> "short", "Medium", "Long", "short", "Long", "-"

```

4.2 Approach 2: Hive Data Processing + WinSCP Export

Process Flow

CSV Dataset → HDFS → Hive → Data Cleaning → WinSCP Export → RStudio Dashboard

Steps:

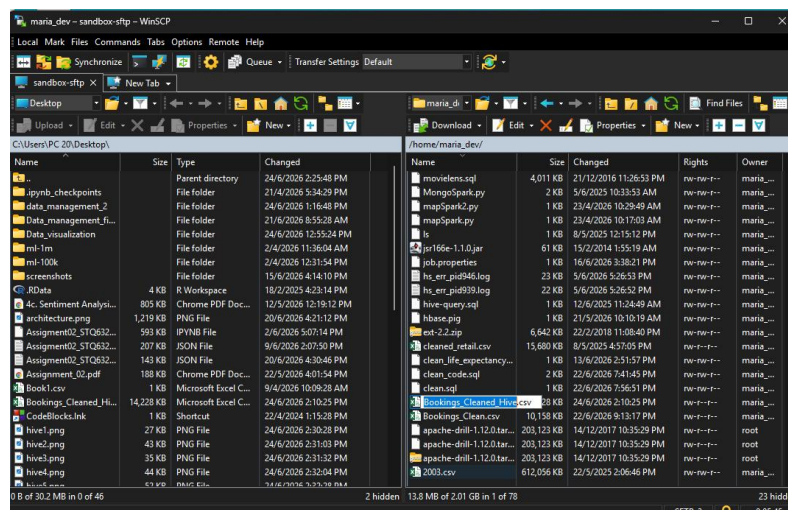
1. Upload CSV file to HDFS.
2. Create Hive table.
3. Execute HiveQL data-cleaning queries.
4. Export processed dataset.
5. Download cleaned data using WinSCP.
6. Load cleaned dataset into RStudio.
7. Build dashboard.

Example Hive Operations:

- NULL handling
- Data transformation
- Derived variable generation
- Aggregation

```
[maria_dev@sandbox-hdp ~]$ hdfs dfs -cat /ride/final/Bookings_Cleaned.Hive.csv | head
1/7/2024 0:00:0:00:00,CNR4091573505,Canceled by Driver,CID742344,eBIke,Yeshwanthpur,Devanahalli,\N,\N,\N,Custom was coughing/sick,\N,\N,367.0,\N,0.0,\N,\N
1/7/2024 0:00:0:00:00,CNR4197668379,Success,CID451984,Prime Plus,Mallleshwaram,BTM Layout,238.0,140.0,\N,\N,No,\N,791.0,Cash,31.0,3.1,4.7
1/7/2024 0:02:0:02:00,CNR7837451859,Canceled by Driver,CID159762,Mini,Padmanabhanagar,Magadi Road,\N,\N,\N,More than permitted people in there,\N,\N,1985.0,\N,0.0,\N,\N
1/7/2024 0:03:0:03:00,CNR3610392570,Success,CID569014,eBIke,JP Nagar,Sarjapur Road,196.0,60.0,\N,\N,Yes,Custom Demand,120.0,Credit Card,26.0,3.1,4.4
1/7/2024 0:04:0:04:00,CNR4163786807,Success,CID967658,Prime Sedan,Hobbal,Vorthur,280.0,130.0,\N,\N,No,\N,914.0,UPI,3.0,4.8,3.3
1/7/2024 0:04:0:04:00,CNR4467034507,Success,CID783274,Prime Plus,Cox Town,Yelahanka,224.0,125.0,\N,\N,No,\N,432.0,Cash,26.0,3.8,3.5
1/7/2024 0:04:0:04:00,CNR8282131599,Success,CID332553,Auto,Langford Town,JP Nagar,259.0,140.0,\N,\N,No,\N,170.0,Cash,5.0,4.3,3.5
1/7/2024 0:04:0:04:00,CNR8299032288,Success,CID874321,Bike,Shivajinagar,Ulsoor,105.0,115.0,\N,\N,No,\N,239.0,Cash,14.0,5.0,3.5
1/7/2024 0:05:0:05:00,CNR2922363339,Canceled by Driver,CID596681,Prime Sedan,RT Nagar,Mysore Road,\N,\N,\N,More than permitted people in there,\N,\N,101.0,\N,0.0,\N,\N
1/7/2024 0:05:0:05:00,CNR2979520808,Success,CID815153,eBIke,Indiranagar,JP Nagar,224.0,60.0,\N,\N,No,\N,127.0,Cash,10.0,3.9,4.1
cat: Unable to write to output stream.
[maria_dev@sandbox-hdp ~]$
```

```
hive> SHOW TABLES;
OK
abc
book1
book1_clean
bookings_raw
employee
flights_2003
j
movie_names
online_retail
ratings
ratings_v2
u
Time taken: 2.035 seconds, Fetched: 12 row(s)
hive>
```



4.3 Approach 3: Direct Hive Connection Using ODBC

Process Flow

CSV Dataset → HDFS → Hive → ODBC Connection → RStudio Dashboard

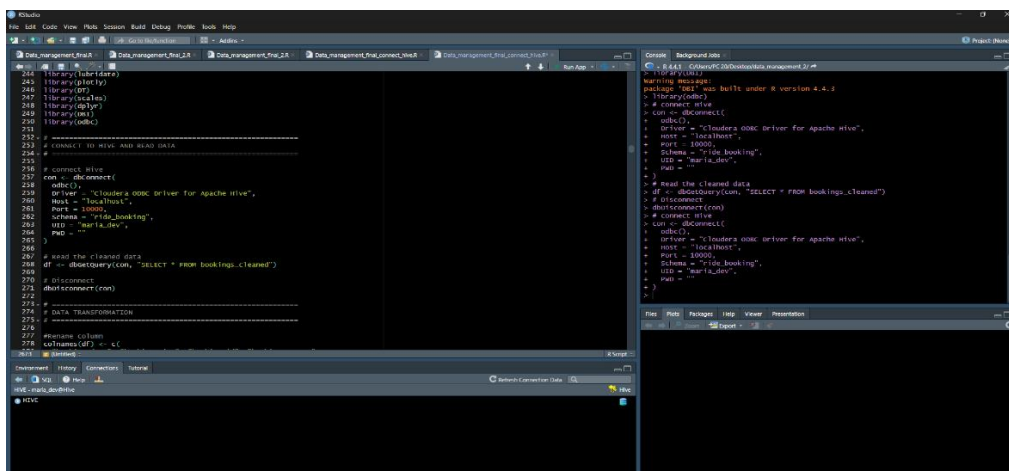
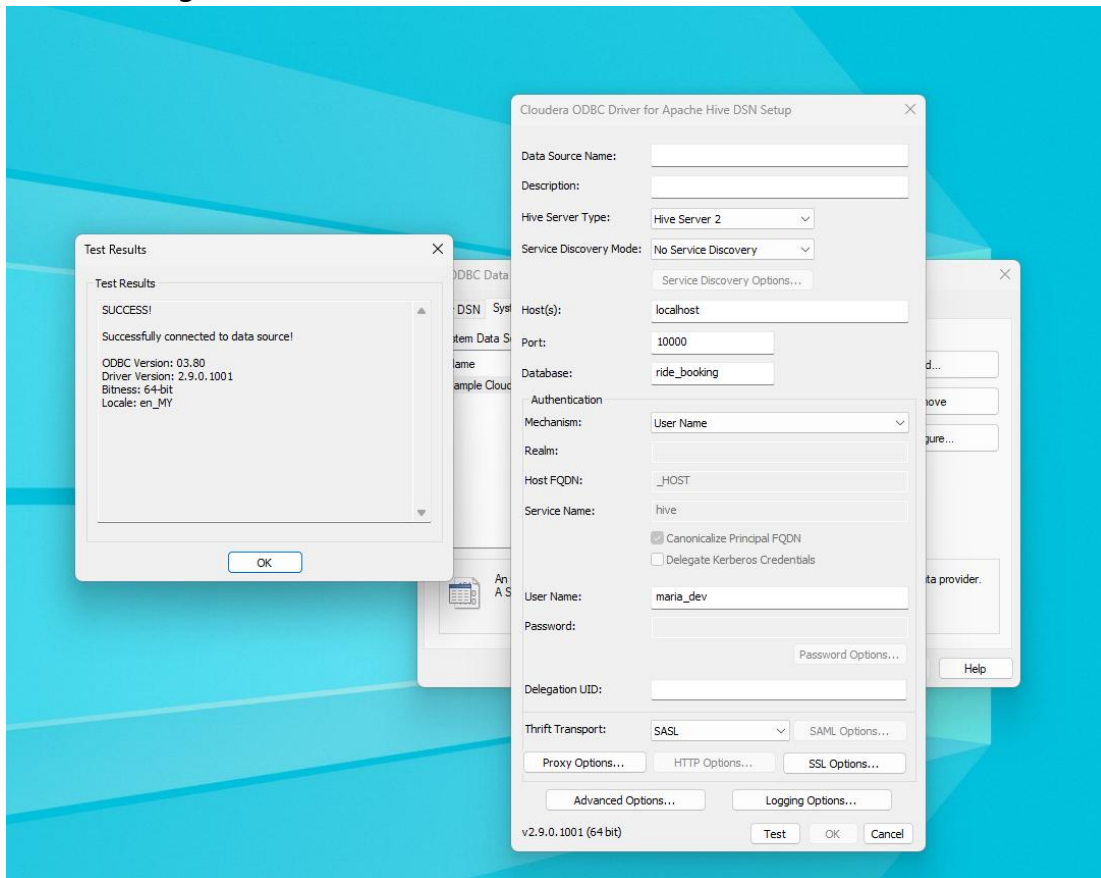
This approach eliminates the need to export files manually.

Steps:

1. Configure Hive ODBC Driver.
2. Create DSN connection.
3. Connect RStudio to Hive.
4. Retrieve Hive tables directly.
5. Build dashboard from live Hive data.

Advantages:

- Real-time access.
- No intermediate CSV files.
- Better integration between Hive and R.



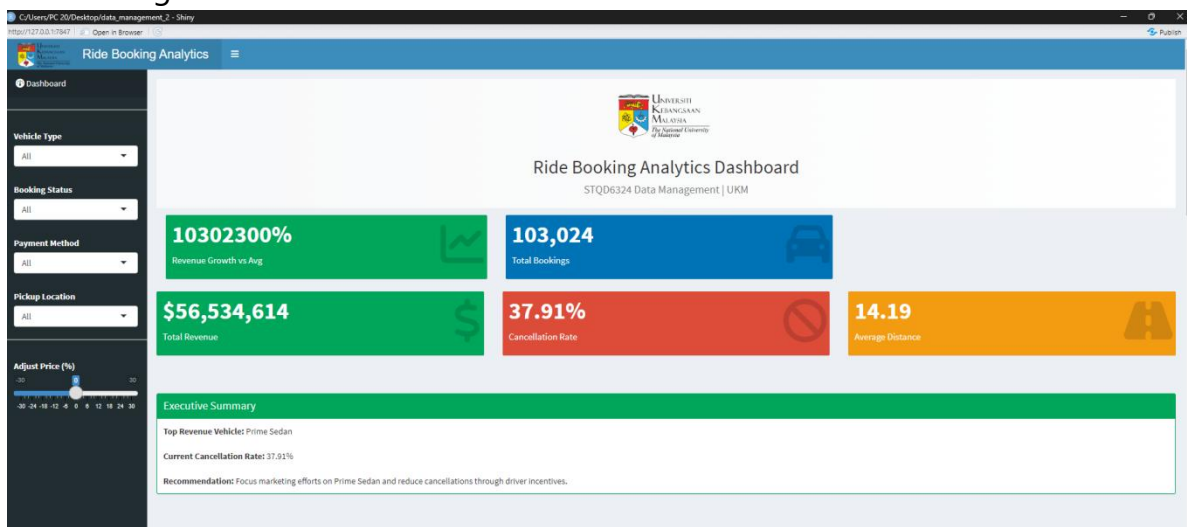
5. Dashboard Development

The dashboard was developed using Shiny Dashboard.

Main features include:

5.1 KPI Monitoring

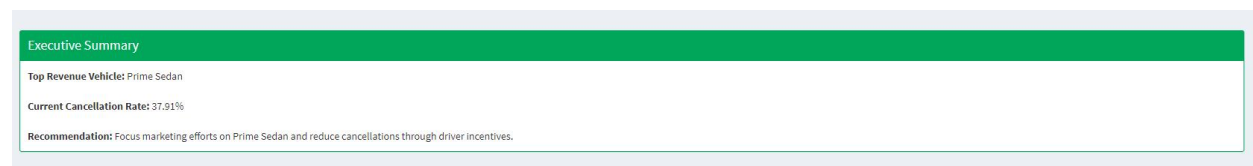
- Total Bookings
- Total Revenue
- Cancellation Rate
- Average Distance



5.2 Executive Summary

Automatically generated summary describing:

- Highest revenue vehicle type
- Current cancellation rate
- Business recommendations



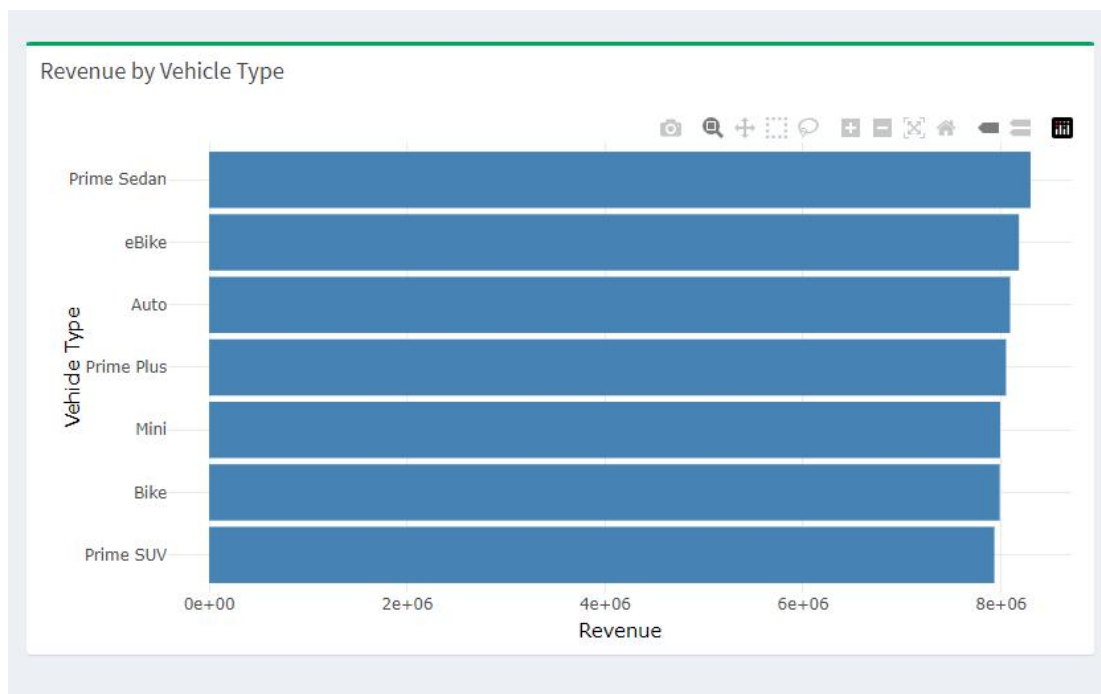
5.3 Booking Trend Analysis

Interactive booking trend visualization over time.



5.4 Revenue Analysis

Revenue distribution by vehicle type.



5.5 Payment Method Analysis

Payment method distribution analysis.



5.6 Cancellation Analysis

Booking status and cancellation analysis.



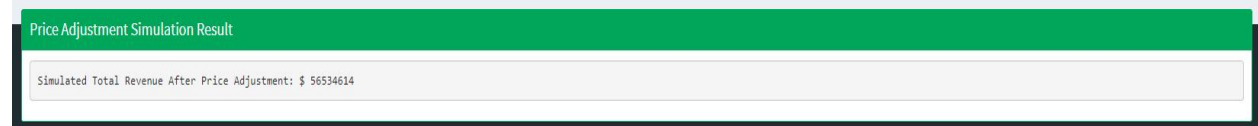
5.7 Ratings Analysis

Customer rating distribution visualization.



5.8 Price Adjustment Simulation

Interactive simulation showing estimated revenue after price adjustments.



5.9 Cancellation Risk Prediction

A logistic regression model was implemented to estimate booking cancellation risk.

Predictor Variables:

- Ride Distance
- Booking Value
- Vehicle Type
- Payment Method

Output:

- Estimated cancellation probability

Cancellation Risk Prediction

Driver Rating: 4.5

Ride Distance (km): 1

Booking Value: 500000

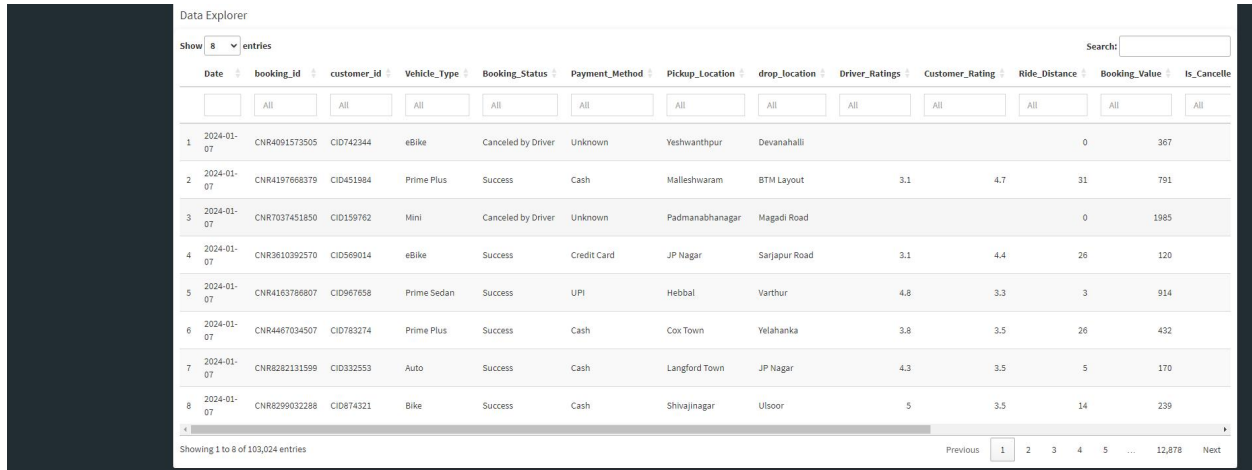
Vehicle Type: eBike

Payment Method: Unknown

22.79% Cancellation Risk

5.10 Data Explorer

Interactive table with filtering functionality.



The screenshot shows a 'Data Explorer' interface. At the top, there's a search bar and a 'show 8 entries' dropdown. Below this is a table with 13 columns: Date, booking_id, customer_id, Vehicle_Type, Booking_Status, Payment_Method, Pickup_Location, drop_location, Driver_Ratings, Customer_Rating, Ride_Distance, Booking_Value, and Is_Canceled. Each column has a filter dropdown set to 'All'. The table contains 8 rows of data, each with a row number and a date. The first row is highlighted. At the bottom, there's a pagination bar showing 'Showing 1 to 8 of 103,024 entries' and a 'Previous' button followed by a series of page numbers (1, 2, 3, 4, 5, ..., 12,878) and a 'Next' button.

	Date	booking_id	customer_id	Vehicle_Type	Booking_Status	Payment_Method	Pickup_Location	drop_location	Driver_Ratings	Customer_Rating	Ride_Distance	Booking_Value	Is_Canceled
1	2024-01-07	CHN4091573505	CID742344	eBike	Canceled by Driver	Unknown	Yeshwanthpur	Devanahalli			0	367	
2	2024-01-07	CHN4197668379	CID451984	Prime Plus	Success	Cash	Malleshwaram	BTM Layout	3.1	4.7	31	791	
3	2024-01-07	CHN7037451850	CID159762	Mini	Canceled by Driver	Unknown	Padmanabhanagar	Magadi Road			0	1985	
4	2024-01-07	CHN3610392570	CID569014	eBike	Success	Credit Card	JP Nagar	Sarjapur Road	3.1	4.4	26	120	
5	2024-01-07	CHN4163766807	CID967658	Prime Sedan	Success	UPI	Hebbal	Varthur	4.8	3.3	3	914	
6	2024-01-07	CHN4467034507	CID783274	Prime Plus	Success	Cash	Cox Town	Yelahanka	3.8	3.5	26	432	
7	2024-01-07	CHN8262131599	CID032553	Auto	Success	Cash	Langford Town	JP Nagar	4.3	3.5	5	170	
8	2024-01-07	CHN8299032288	CID874321	Bike	Success	Cash	Shivajinagar	Ulsoor	5	3.5	14	239	

6. Results and Discussion

The dashboard successfully provides business insights regarding:

- Revenue performance
- Customer booking behaviour
- Cancellation trends
- Vehicle usage patterns

Among the three approaches:

Approach	Advantages	Limitations
R Only	Simple implementation	Limited scalability
Hive + WinSCP	Better data management	Requires manual export
Hive + ODBC	Real-time integration	More complex configuration

The Hive + ODBC approach provides the most efficient and scalable solution.

7. Conclusion

This project successfully demonstrates the integration of Apache Hive and RStudio for ride-booking analytics.

Three different implementation approaches were evaluated:

1. Direct R processing.
2. Hive processing with exported datasets.
3. Direct Hive-ODBC integration.

The developed dashboard enables interactive analysis, business reporting, and predictive analytics. The project highlights the importance of data management technologies in supporting modern business intelligence applications.

References

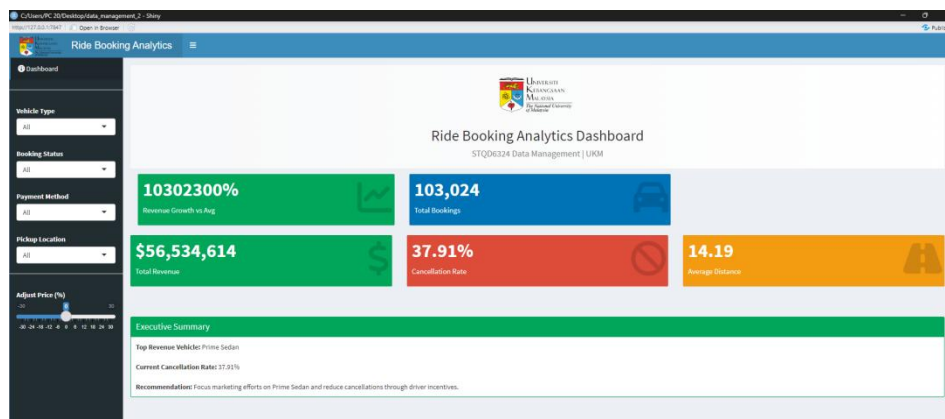
Apache Hive Documentation

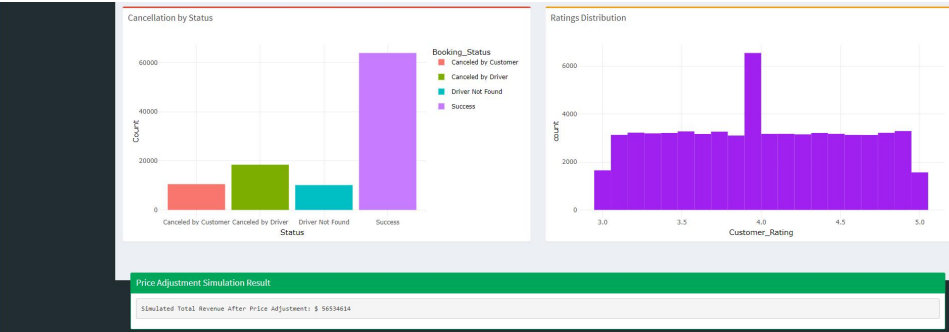
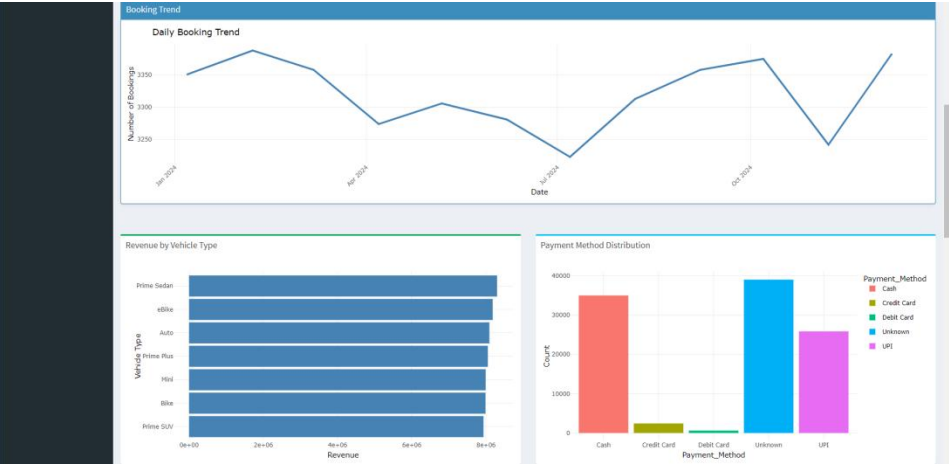
R Project Documentation

Shiny Documentation

ODBC Documentation

UKM STQD6324 Data Management Course Materials





Cancellation Risk Prediction

Driver Rating: 4.3, Ride Distance (km): 33, Booking Value: 500, Vehicle Type: eBike, Payment Method: Unknown

0% Cancellation Risk

Data Explorer

Show 8 entries

	Date	booking_id	customer_id	Vehicle_Type	Booking_Status	Payment_Method	Pickup_Location	drop_location	Driver_Ratings	Customer_Rating	Ride_Distance	Booking_Value	Is_Canceled
1	2024-05-07	CNR4011573505	CID742344	eBike	Cancelled by Driver	Unknown	Nehruviharipur	Devanahalli			0	367	
2	2024-05-07	CNR419766379	CID453184	Prime Plus	Success	Cash	Malleshwaram	BTM Layout	3.1	4.7	31	791	
3	2024-05-07	CNR7037451850	CID159762	Mini	Cancelled by Driver	Unknown	Padmanabhanagar	Magadi Road			0	1365	
4	2024-05-07	CNR3610302570	CID568054	eBike	Success	Credit Card	JP Nagar	Sarjapur Road	3.1	4.4	26	120	
5	2024-05-07	CNR4163786807	CID947658	Prime Sedan	Success	UPI	Hebbal	Varthur	4.8	3.3	3	914	
6	2024-05-07	CNR4467034507	CID763274	Prime Plus	Success	Cash	Cox Town	Yelahanka	3.8	3.5	26	432	
7	2024-05-07	CNR8182131599	CID332553	Auto	Success	Cash	Langford Town	JP Nagar	4.3	3.5	5	170	
8	2024-05-07	CNR8296032288	CID874321	Bike	Success	Cash	Shivajinagar	Uttoor	5	3.5	14	239	

Showing 1 to 8 of 103,024 entries